“Clothing Brand Web System for Online Retail Operations”

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Approval Sheet

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SOFTWARE DESIGN PROJECT INFORMATION

2nd Semester, SY 2025-2026

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Project Title	Clothing Brand Web System for Online Retail Operations
Project Concentration Area	Web Systems, E-Commerce Technologies, Database-Driven Applications
Design Objectives	The general objective is to provide a comprehensive design and development plan for the clothing brand web system. This includes the features and website functions that will be implemented in the project. Specifically, it aims to: 1. Develop a Web Based system that; • That showcase the clothing (Hoodies, Polo, Pant etc..) of Doing Good • Shows the categories of clothe based gender • Has a database management for products, users, orders, and inventory 2. Test and evaluate the system's accuracy and performance
Constraints	
Performance Constraints (Response Time)	Performance constraint relates to the speed and responsiveness of the website, measured by load and response time. Poor performance reduces user satisfaction. Therefore, faster systems are more effective. Hence, the design with the best performance is considered the best design.

Security Constraints (Data Protection Level)	Security constraint relates to protecting user data and system integrity through secure methods. Weak security increases the risk of data breaches. Therefore, stronger security is more reliable. Hence, the design with the highest security is considered the best design.
Scalability Constraints (Number of Concurrent Users)	Scalability constraint relates to the system's ability to handle growth in users and data. Poor scalability leads to slowdowns or failures. Therefore, scalable systems are more reliable. Hence, the design with the highest scalability is considered the best design.
Efficiency (Inference Time)	Efficiency constraint relates to the optimal use of resources such as time, cost, and system power. Inefficiency increases costs and slows performance. Therefore, efficient systems are more effective. Hence, the design with the highest efficiency is considered the best design.
Technology Constraints (Technology Stack)	Technology constraint relates to the tools and platforms used in development. Poor technology choices can limit performance and maintenance. Therefore, modern and compatible technologies are more effective. Hence, the design with the best technology is considered the best design.
Other constraints: These constraints do not affect each design; therefore, these were not included in selecting the best design.	
Public Health	Websites contribute to public health by sharing reliable information, promoting safe online practices, and protecting user data privacy. They can also support awareness campaigns and provide access to important services.
Welfare	Websites improve user welfare by offering convenient access to products and services, secure transactions, and efficient customer support, making everyday activities easier and more accessible.
Social	Websites strengthen social connections by enabling communication, interaction, and community building through online platforms, feedback systems, and shared content.
Global	Websites allow businesses and organizations to reach international audiences, operate across different regions, and serve users worldwide without geographical limitations.
Cultural	Websites can respect and represent different cultures by providing multilingual support, inclusive design, and content that considers cultural values, traditions, and diversity

Standards	
Standard A - WCAG 2.1 Level AA	This standard makes sure websites are accessible to people with disabilities. In the project, it was used to improve text contrast, add text to product images, and ensure the site can be used with a keyboard.
Standard B - OWASP Top 10	This standard lists the most common security risks for web systems. In the project, it helped guide secure login, protect user data, and prevent issues like unsafe input or weak passwords.

Abstract

The design and development of a web system for a clothing brand that will facilitate online retail operations is presented in this project. The system's main goals are to create an intuitive user interface for administrators and customers alike, efficiently arrange product catalogs, and create a seamless inventory and order management workflow. A database system to manage products, users, orders, and stock levels, as well as front-end modules for user interaction and back-end modules for data processing, are all part of the project. Performance testing and functional assessments were carried out to confirm accuracy, responsiveness, and dependability in order to guarantee system quality. To improve user experience and safeguard sensitive data, accessibility and security standards, such as WCAG 2.1 Level AA and the OWASP Top 10, were implemented. The final design provides a scalable, safe, and effective web-based retail solution that is suited to a contemporary clothing brand's requirement.

Keywords: Web System Design, Online Retail, Clothing Brand, User Interface, Product Catalog Management, Inventory and Order Management, Front-End Development, Back-End Development, Database Management, System Performance, Accessibility Standards, WCAG 2.1 Level AA, Security Standards, OWASP Top 10, Scalable Web Solution.

List of Tables

List of figures

List of abbreviation

Definition of terms

TABLE OF CONTENTS

SOFTWARE DESIGN TITLE ..... **Error! Bookmark not defined.**
Approval Sheet 2
Abstract 6
List of Tables 7
List of figures 8
List of abbreviation 9
Definition of terms 10
TABLE OF CONTENTS 11
CHAPTER 1: THE PROJECT AND ITS BACKGROUND 14
1.1 The Problem 14
1.2 The Client 14
1.3 The Project 15
1.4 Project Objectives 16
1.5 Scope and Delimitations 16
1.6 Design Constraints 17
1.7 Engineering Standards 18
1.8 Engineering Design Process 18
CHAPTER 2: SOFTWARE DESIGN 23
2.1 Description of the Design Solution 23
2.1.1 General Description 23
2.1.2 Engineering Principles Involved 23
2.1.3 Prior Art Analysis 26
2.2 General System Architecture 28
2.2.1 Hardware Elements 28
2.2.2 Software Elements 28
2.2.2 System Algorithm 29
2.2.3 Data, Datasets, and Processing 29
2.3 Design Alternatives 30
2.3.1 Rationale for Design Alternatives 30
2.3.1 Design Alternative 1: 31
2.3.2 Design Alternative B 32
2.3.3 Design Alternative C 33
2.4 Standards Involved in the Design 33

CHAPTER 3: DESIGN TRADEOFFS 33
3.1 Summary of Constraints 36
3.2 Trade-offs 36
3.2.1 Tradeoff 1: Constraint A (Metric) 37
3.2.2 Tradeoff 2: Constraint B (Metric) 38
3.2.3 Tradeoff 3: Constraint C (Metric) 38
3.2.4 Tradeoff 4: Constraint D (Metric) 39
3.2.5 Tradeoff 5: Constraint E (Metric) 39
3.3 Summary of the Normalized Values of the Three Designs 40
3.4 Designers Raw Ranking for the Three Designs 40
3.5 Sensitivity Analysis 40
3.6 Influence of the Design Tradeoffs in the Final Design 40
CHAPTER 4: FINAL DESIGN 41
4.1 Final Design 41
4.1.1 Software Application 41
4.1.2 Hardware Design 41
4.2 Test Procedures and Evaluation 41
4.2.1 Test Procedures 41
4.2.2 Test Evaluation 41
4.3 Test and Evaluation Results 41
4.3.1 Test Results 41
4.3.2 Evaluation Results 41
4.4 Conclusion 41
4.5 Impact of the Design 41
4.5.1 Societal 41
4.5.2 Ethical 41
4.5.3 Legal 41
4.6 Sustainability Plan 41
CHAPTER 5: BUSINESS PLAN AND MODEL 42
5.1 Business Plan 42
5.1.1 Executive Summary 42
5.1.2 General Company Description 42
5.1.3 Products and Services Offered 42
5.1.4 Marketing Plan 42

5.1.5 Marketing Strategy 42
5.2 Business Model 42
5.3 Intellectual Property (IP) Reports 42
REFERENCES 43
APPENDICES 44

CHAPTER 1: THE PROJECT AND ITS BACKGROUND

This chapter highlights the need for a web system to help the clothing brand improve visibility, manage inventory efficiently, and increase sales. By applying engineering standards and a structured design process, the project ensures the system is secure, reliable, and aligned with the client's goals.

1.1 The Problem

The Doing Good Clothing business faces several challenges that limit its growth and market reach, particularly in the absence of a dedicated online platform. One of the primary concerns is the inability to expand the brand name globally, as the business currently relies on limited channels that restrict its visibility to a wider audience. This lack of digital presence prevents the business from competing effectively in the growing online clothing market.

Low sales have also been observed due to insufficient online visibility, making it difficult for potential customers to discover the products and services offered. Without a centralized platform, customers are unable to access important information about the business at any time, which affects convenience and reduces engagement. This limitation can result in missed opportunities for both customer acquisition and retention.

Furthermore, the absence of an efficient system to showcase products, prices, and updates creates a gap in communication between the business and its customers. Traditional methods of promotion are often limited and inconsistent, making it harder to maintain customer interest and trust. Therefore, there is a need for a well-designed website that can provide accessible, organized, and up-to-date information.

With the advancement of web technologies and e-commerce platforms, developing a website presents a practical solution to these challenges. A dedicated online system can enhance visibility, improve customer experience, and support business growth by enabling global reach and efficient product presentation.

1.2 The Client

The client for this project is Mr. Dexter Luis Vega, the owner of an established clothing brand that is currently operating without its own online retail system. To support the growth and digital expansion of his business, our team will be developing a web-based platform that aligns with the brand's operational needs and customer expectations.

Through the interview and data-gathering process, our team will identify the specific requirements of Mr. Vega's clothing brand. The planned system will provide structured product displays, real-time inventory management, secure customer transactions, and an organized order processing workflow. By creating this platform, the project will contribute to enhancing the brand's online presence, accessibility, and overall retail efficiency.

Table 1-1 presents the main client and engineering requirements for the Doing Good Clothing website system. The client requirements highlight the business needs, such as enabling customers to easily browse products, ensuring secure handling of user accounts and transactions, and maintaining accurate inventory tracking. These reflect the operational challenges of improving customer experience, increasing online visibility, and managing product information efficiently.
Each engineering requirement is designed to be specific and actionable, translating these business needs into technical solutions. This includes implementing a responsive and accessible website design, applying secure authentication and data protection practices, and developing a database-driven inventory system supported by backend technologies such as Node.js. These considerations ensure that the website is user-friendly, secure, and capable of supporting business growth and efficient operations.

Table 1-1. Client and Engineering Requirements / Considerations

Client Requirements / Considerations	Engineering Requirements / Considerations
The website can allow customers to browse and view clothing products easily.	The website should have a responsive design, fast loading, organized navigation, and comply with WCAG 2.1 Level AA to ensure accessible browsing.
The website can securely store user accounts and handle online transactions safely.	Implement secure authentication, encrypted data storage, and OWASP Top 10 security practices.
The website can track inventory and update stock levels automatically when orders are placed.	Create a database with automated inventory management logic using backend APIs in Node.js.

1.3 The Project

This project focuses on creating a web system for Mr. Dexter Luis Vega's clothing brand to help the business operate online. The system will allow customers to browse products and make purchases, while administrators can manage products, inventory, and orders efficiently. The system will have a front-end for users to interact with, a back end to process orders and update data, and a database to store product, user, and sales information. It will also follow accessibility and security standards to ensure it is safe and easy to use. By developing this system, our team will help the brand expand online, improve operations, and prepare for future growth.

1.4 Project Objectives

The primary objective of this project is to design and develop a comprehensive website to facilitate the online retail operations of Mr. Dexter Luis Vega's clothing brand. The system will integrate functionalities for efficient management of product catalogs, inventory control, order processing, and user interaction.

Specifically, the project aims to:

1. Develop software modules encompassing both front-end and back-end functionalities, including the following:

- dynamic product display
- secure transaction handling
- database management
- real-time inventory synchronization

2. Test and evaluate

- the system's accuracy
- performance
- security
- reliability

1.5 Scope and Delimitations

This project focuses on designing and developing a web system for Mr. Dexter Luis Vega's clothing brand to support online sales and business operations. The system will include features for managing product listings, tracking inventory, processing customer orders, and handling user accounts for both customers and administrators. The system will follow important standards to ensure it is accessible and secure.

The web system will be built using React for the user interface and Node.js for the server side, along with a database to store product and customer data. The system will be tested in controlled environments to check its accuracy and performance under normal use. This project covers only the web application and does not include developing a mobile app or connecting with external delivery or payment services beyond basic online transactions. It will support up to 1,000 users at the same time but will not include advanced performance tuning for larger traffic. Security will protect against common threats, but highly advanced cyberattacks are beyond the project's scope. Marketing, sourcing of products, and physical inventory handling outside the system are not part of this project. The user interface will focus on core functions without extensive design customization or branding details.

1.6 Design Constraints

Several constraints were considered when planning the project, and these constraints limit the system's technological features while also aiding in the refinement of design possibilities during development. However, only the top five limitations were applied in the project's development process because the remaining constraints had no direct impact on the project.

Performance Constraints (Response Time)

Performance constraint relates to the speed and responsiveness of the website, measured by load and response time. Poor performance reduces user satisfaction. Therefore, faster systems are more effective. Hence, the design with the best performance is considered the best design.

Security Constraints (Data Protection Level)

Security constraint relates to protecting user data and system integrity through secure methods. Weak security increases the risk of data breaches. Therefore, stronger security is more reliable. Hence, the design with the highest security is considered the best design.

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Technology constraint relates to the tools and platforms used in development. Poor technology choices can limit performance and maintenance. Therefore, modern and compatible technologies are more effective. Hence, the design with the best technology is considered the best design.

Other constraints: These constraints do not affect each design; therefore, these were not included in selecting the best design.

Public Health

Websites contribute to public health by sharing reliable information, promoting safe online practices, and protecting user data privacy. They can also support awareness campaigns and provide access to important services.

Welfare

Websites improve user welfare by offering convenient access to products and services, secure transactions, and efficient customer support, making everyday activities easier and more accessible.

Social

Websites strengthen social connections by enabling communication, interaction, and community building through online platforms, feedback systems, and shared content.

Global

Websites allow businesses and organizations to reach international audiences, operate across different regions, and serve users worldwide without geographical limitations.

Cultural

Websites can respect and represent different cultures by providing multilingual support, inclusive design, and content that considers cultural values, traditions, and diversity

1.7 Engineering Standards

The engineering standards used in this project ensure that the clothing brand web system is accessible, secure, reliable, and aligned with accepted practices for modern web applications. These standards support the system's design goals and help maintain quality throughout development.

WCAG 2.1 Level AA (Web Accessibility Standard)

This standard ensures that the website is accessible to all users, including those with disabilities. It supports the project's goal of creating an easy-to-use interface by guiding text contrast, keyboard navigation, readable layouts, and alt-text for product images. This directly strengthens the system's usability requirement.

OWASP Top 10 (Web Security Standard)

This standard lists the most common web security risks. It connects to the project's constraint on data protection by helping identify and prevent issues such as weak authentication, unsafe input handling, and data exposure. This ensures that user accounts, transactions, and stored data remain secure.

ISO/IEC 25010 (Software Quality Standard)

This standard guides how software quality is evaluated. It supports the project's testing requirements by defining expectations for performance, reliability, usability, and security. This connects to your objective of ensuring that "all functions operate correctly" and the system responds efficiently.

ISO/IEC 27001 (Information Security Standard)

This standard provides principles for protecting sensitive information. It supports the project's need for secure data storage and processing, especially for customer information, orders, and administrator accounts. It directly connects to your security constraints and database requirements.

ISO/IEC 14543-3 (System Architecture Standard)

Although originally intended for electronic systems, this standard is used here to support a clear and organized system architecture. It helps guide how system components communicate and work together, which connects to your objective of designing the overall structure of the system (front-end, back-end, and database).

1.8 Engineering Design Process

The image shows two people wearing stylish and comfortable clothes. Just like how clothing is planned and created, the Engineering Design Process also follows steps to make a good final product. It starts with finding a problem, thinking of ideas, testing them, and improving the design. This process helps make sure the result is useful, creative, and fits what people need.

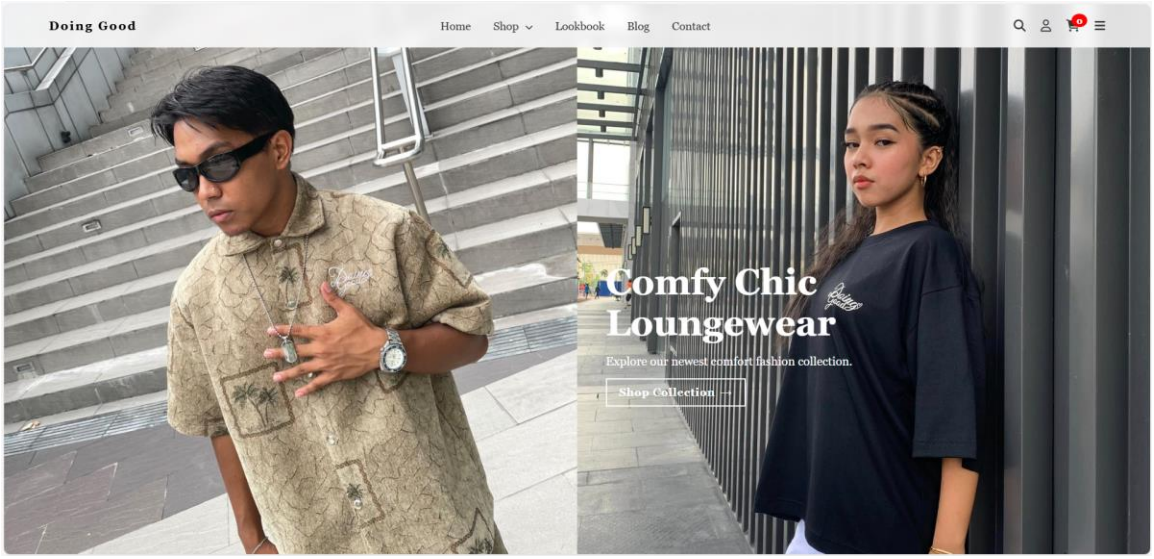

Figure 1.1 The Engineering Design Process (Teach Engineering, 2025)

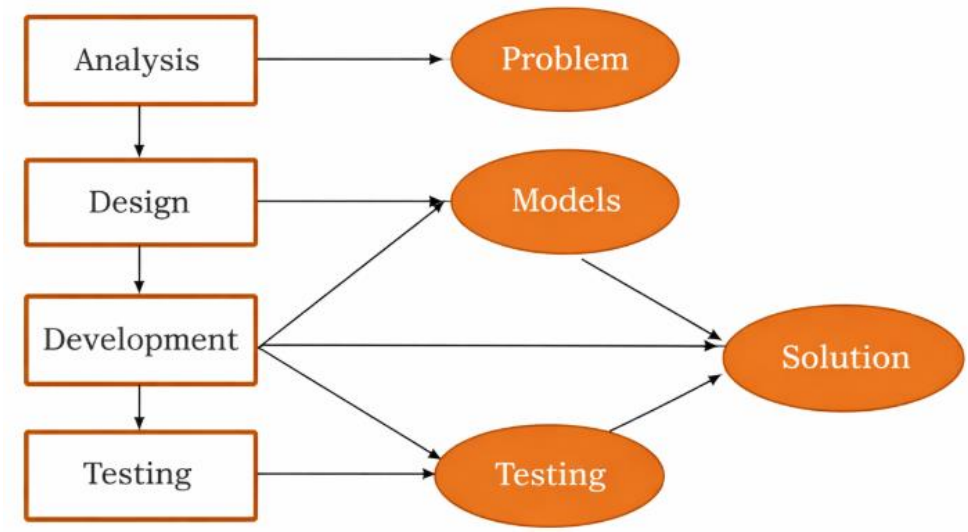

Figure 1.2 The Engineering Design Process (TeachEngineering, 2025)

The picture depicts two individuals dressed comfortably and stylishly. The Engineering Design Process follows steps to produce a high-quality final product, just like clothing is planned and made. These steps are clearly depicted in the diagram. Analysis is the first step, where we comprehend the issue or requirement. After that, we proceed to the Design stage, where concepts and models are developed. These concepts are then transformed into actual products during the development stage. Lastly, testing verifies that everything functions properly. These procedures aid in enhancing the design and guarantee that the finished product is practical, imaginative, and meets people's needs.

1.8.1 Ask: Identifying the Need and Constraints

This step entails figuring out what the client needs and comprehending the main issue. A fully functional clothing brand web system that facilitates online retail operations is required for this project. Constraints like performance limitations, security needs, financial limitations, and accessibility requirements are identified by the team. By posing crucial queries, such as "What should the system do?" Who is going to use it? What limits must it follow? —the group creates a strong basis for the design.

1.8.2 Research the Problem

The team collects pertinent data and examines current systems after determining the need. This entails examining industry standards like WCAG 2.1 and OWASP Top 10, evaluating existing e-commerce platforms, and conducting client interviews to gain a deeper understanding of operational requirements. Technologies like React, Node.js, and database structures are also covered in research. This stage guarantees that the project is based on well-informed choices rather than conjecture.

1.8.3 Imagine: Develop Possible Solution

The group imagines and brainstorms different solutions in this step. This could include different database designs, inventory tracking strategies, and user interface layouts for the web system. Here, the team is encouraged to be creative and can freely explore ideas without worrying about constraints right away. To determine the most promising system structure, a number of design options are drafted.

1.8.4 Plan: Select a Promising Solution

We brainstormed several methods the company can use to help customers explore and purchase clothing through an effective clothing business website. Since the business already has images available from the client's Facebook page, these can be used as visual content for the website.

Option A: A gallery-style website that highlights "Fashion Inspiration" photos taken from the client's Facebook page. This approach showcases different outfits and clothing items in a visually appealing way to inspire customers and attract their attention.
Option B: A product-focused website that provides detailed information for each clothing item, such as available sizes, color options, price, and care instructions, helping customers make informed purchasing decisions.

Option C: A combination approach that uses images from the Facebook page while organizing the products into clear categories such as Tops, and Bottoms, this allows customers to both view attractive clothing photos and easily navigate through the product categories.

1.8.5 Create: Build a Prototype

A working model or preliminary version of the web system is created during this stage. Simple versions of the front-end and back-end interactions, the basic UI framework, or the initial database setup are examples of prototypes. The objective is to develop a working prototype that shows how the finished system will function, not to achieve perfection. Early on in the development process, this prototype aids in identifying possible problems.

1.8.6 Test and Evaluate the Prototype

Once the prototype is created, it undergoes performance, usability, and security testing. The team examines whether the system meets the client's requirements and satisfies constraints such as response time, accessibility, and data protection. User feedback and testing tools help uncover errors, weaknesses, and areas for improvement. Evaluation ensures that the design functions reliably before moving to full development.

1.8.7 Improve: Redesign as Needed

Based on test results, the team revises and enhances the design. Improvements may involve fixing bugs, optimizing performance, strengthening security features, or reorganizing the interface for better user experience. This step reflects the iterative nature of engineering: the design is refined repeatedly until it meets the required standards and delivers a high-quality final pro

CHAPTER 2: SOFTWARE DESIGN

This chapter presents the software design of the proposed Clothing Brand Web System for Online Retail Operations. It outlines the design solution, system architecture, data management approach, applied engineering principles, and the design alternatives evaluated prior to selecting the final implementation. This chapter aims to clearly describe the system structure and demonstrate how the design meets the project objectives and constraints defined in Chapter 1.

2.1 Description of the Design Solution

2.1.1 General Description

The proposed design solution is a web-based e-commerce system that allows customers to browse products, create accounts, place orders, and track purchases, while enabling administrators to securely manage products, inventory, and customer orders through a centralized dashboard.

The system follows a three-tier architecture:

1. **Presentation Layer (Front-end)** – built using React, responsible for user interaction and interface rendering.

2. **Application Layer (Back-end)** – developed using Node.js and Express, responsible for business logic, authentication, order processing, and inventory updates.

3. **Data Layer (Database)** – stores product information, user accounts, orders, and inventory records.

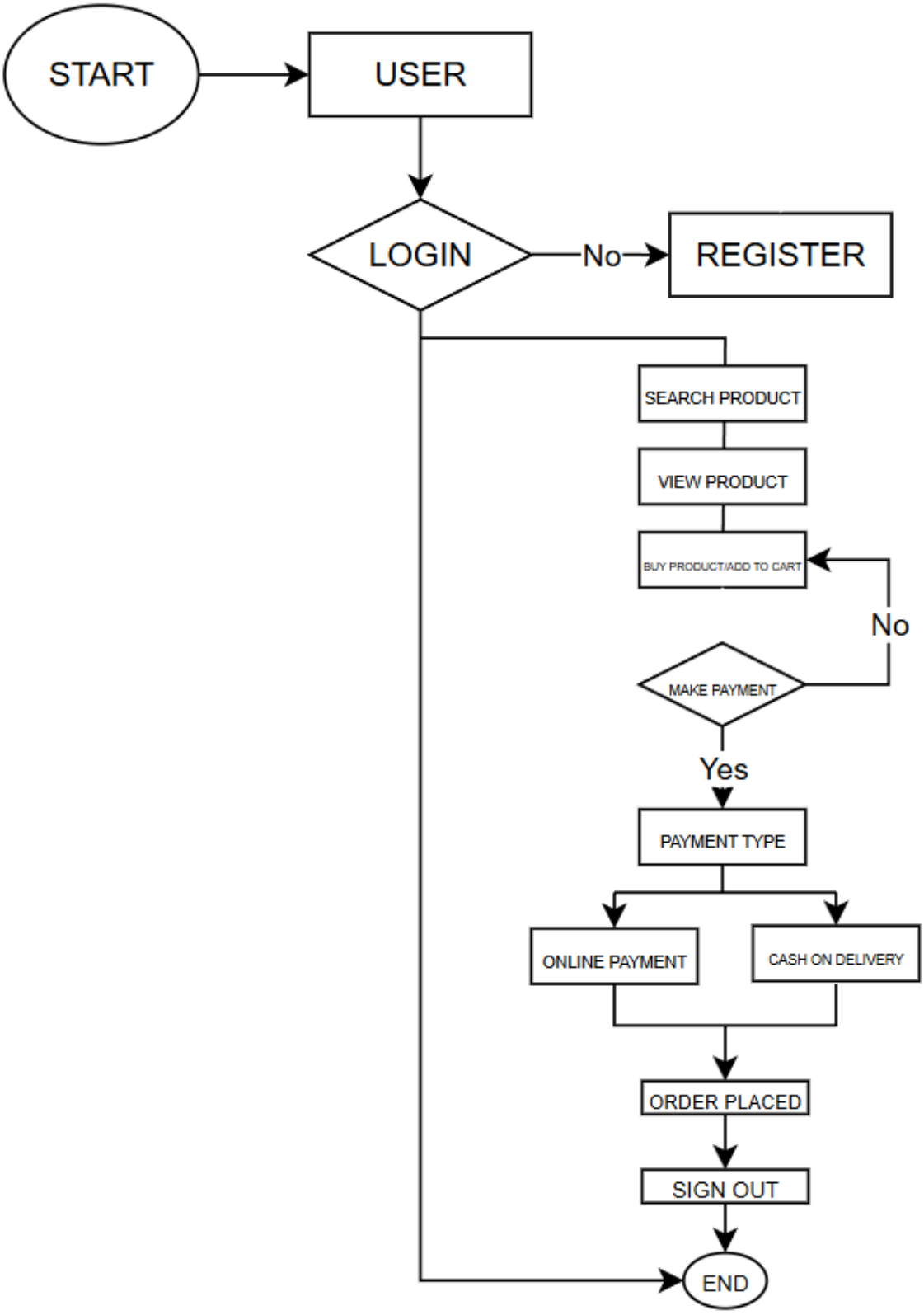

Figure 2.1 Illustrative Diagram of the System

Users access the system through a web browser, where requests are handled by the React-based front end and forwarded to the Node.js API server. The server processes each request, retrieves or updates data from the database, and returns the appropriate response to the front end for display's Administrators follow the same process but are granted access to additional system management modules.

This architecture promotes separation of concerns, simplifies system maintenance, supports scalability, and enhances overall security.

2.1.2 Engineering Principles Involved

The system design applies several fundamental software and systems engineering principles to ensure reliability, maintainability, and scalability:

1. Modularity

- The system is structured into independent modules, including authentication, product management, order processing, and inventory control. This modular approach simplifies debugging, testing, and future enhancements.

2. Abstraction

- Complex operations such as payment processing, database interactions, and security validation are encapsulated within application programming interfaces (APIs) and service layers, allowing developers to work with simplified and consistent interfaces.

3. Separation of Concerns

- The system architecture clearly separates responsibilities among the front end (presentation), back end (business logic), and database (data storage). This improves maintainability and reduces overall system complexity.

4. Scalability

- The client–server architecture enables the system to accommodate increasing user demand by supporting server upgrades and the integration of load-balancing mechanisms when necessary.

5. Reliability and Fault Tolerance

- Input validation, structured error handling, and logging mechanisms are implemented to ensure stable operation and facilitate rapid identification and resolution of system issues.

6. Security by Design

- Security considerations are incorporated from the initial design phase, including encrypted password storage, role-based access control, and secured API endpoints to protect sensitive data and system resources.

2.1.3 Prior Art Analysis

Several existing e-commerce systems influenced the design of this project, particularly the official Nike website, which serves as a benchmark for modern online retail platforms. The Nike website demonstrates effective product presentation, streamlined navigation, secure user account management, and efficient order processing workflows.

Narrative Comparison:

The Nike website provides a highly responsive and visually structured product catalog that enhances user experience and supports quick product discovery. It also features advanced filtering options, real-time inventory visibility, and an intuitive checkout process that minimizes user effort.

In addition, the platform integrates secure authentication mechanisms and order tracking functionalities, allowing customers to monitor their purchases seamlessly. However, such enterprise level systems typically require significant development resources and infrastructure, making them less practical for small to medium-sized businesses.

In contrast, the proposed system adopts similar functional concepts and usability principles while maintaining a simplified architecture and lower implementation cost, making it more suitable for the client's operational scale and budget constraints.

Matrix Format is preferred, in addition to narratives. Explain the table.

Table 2.1 Prior Art Analysis Matrix

Design	Features				
	Product Catalog Organization	**Advance Search & Filtering**	**User Account Management**	**Order Tracking**	**Brand Customization**
Nike Website	x	x	x	x	
Zara Website	x		x	x	
Brandy Melville Website	x			x	
Urban Outfitters Website	x		x	x	
Generic E-commerce Template	x		x	x	
Proposed System (Doing Good)	x	x	x	x	x

Table 2.1 presents the Prior Art Analysis Matrix for the Doing Good Clothing website, comparing existing e-commerce platforms with the proposed system based on key features such as product catalog organization, advanced search and filtering, user account management, order tracking, and brand customization, where each check mark (x) indicates the presence of a feature. While established platforms like Nike and Zara include several core features, and others such as Brandy Melville, Urban Outfitters, and generic e-commerce templates provide only partial functionality, these systems often lack a complete integration of all essential and advanced features. In contrast, the proposed Doing Good Clothing website stands out by incorporating all key functionalities, including advanced search capabilities and full brand customization, which are not consistently present in the compared systems. This comprehensive approach ensures a more personalized, user-friendly, and efficient shopping experience. Overall, the table highlights the limitations of existing platforms and emphasizes the proposed system as a more innovative and competitive solution in the e-commerce space.

2.2 General System Architecture

2.2.1 Hardware Elements

The proposed system does not require specialized or dedicated hardware components. It is a purely software-based web application that operates using existing client devices and standard server infrastructure. Users access the system through commonly available devices such as desktop computers, laptops, tablets, or smartphones with web browsers. Server-side operations are handled using standard web hosting environments without the need for custom hardware configurations.

2.2.2 Software Elements

A. Embedded Software

Web-Based System, no embedded software is required for this project, as the Doing Good Clothing system is purely web-based and does not rely on hardware components or physical devices. Unlike systems that require microcontrollers or embedded processors, this platform operates entirely through internet browsers and server-side infrastructure. All functionalities, including data processing, user interaction, and system operations, are handled through web technologies, making the system more accessible, scalable, and easier to maintain.

B. Application Software

React, HTML, CSS, and JavaScript

The frontend of the system is developed using React along with core web technologies such as HTML, CSS, and JavaScript. React enables the creation of dynamic and responsive user interfaces, allowing users to browse products, manage accounts, and interact with the website efficiently. These technologies ensure a smooth and visually appealing user experience across different devices and screen sizes.

Node.js and Express.js

The backend of the system is implemented using Node.js and Express.js, which handle server-side logic, API integration, and communication between the frontend and database. These technologies enable efficient handling of user requests, order processing, and system operations, ensuring fast and reliable performance of the web application.

MySQL / PostgreSQL

A relational database such as MySQL or PostgreSQL is used to store and manage system data, including user accounts, product information, orders, and inventory. This ensures data consistency, structured storage, and efficient retrieval of information necessary for system functionality.

JWT and bcrypt

Security within the system is implemented using JSON Web Tokens (JWT) for authentication and bcrypt for password hashing. JWT ensures secure user sessions and access control, while bcrypt protects user passwords by encrypting them before storage. These tools are essential in maintaining data privacy and system security.

Both customer and administrator interfaces are browser-based, eliminating the need for additional software installation and allowing easy access to the system.

C. Key Algorithms Used

User Authentication and Authorization Algorithms

The system uses authentication and authorization algorithms to verify user identities and control access to system features. These algorithms ensure that only authorized users can access sensitive data and perform specific actions within the platform.

Inventory Synchronization Logic

Inventory synchronization logic is implemented to automatically update product stock levels whenever transactions occur. This ensures real-time accuracy of inventory data and prevents issues such as overselling or stock inconsistencies.

Order Validation and Processing Algorithms

Order validation and processing algorithms are used to verify customer orders, ensure correct data input, and manage the complete transaction process. These algorithms help maintain system reliability and ensure that orders are processed efficiently and accurately.

2.2.2 System Algorithm

The system follows a structured request–response algorithm:

1. User submits a request through the frontend interface.
2. The frontend validates input and sends the request to the backend API.
3. The backend authenticates the user and processes business logic.
4. The database is queried or updated as required.
5. The backend sends a response to the frontend.
6. The frontend displays the result to the user.

2.2.3 Data, Datasets, and Processing

A. Datasets

The system utilizes datasets that are primarily generated internally and partially collected from existing business sources. The data used in the Doing Good Clothing web system includes product data such as clothing names, categories, sizes, prices, stock levels, and images, as well as user data containing customer and administrator information like names, email addresses, and passwords. In addition, order data is also included, which consists of transaction details such as selected items, quantities, order status, and payment information.

The data acquisition process begins with manual data entry, where the administrator inputs initial product information into the system based on the available inventory. Some data, such as product images and descriptions, are collected from existing sources like the business's Facebook page and previous listings. Furthermore, user-generated data is dynamically collected when customers register, log in, and place orders on the website. All collected data is then stored and managed in a MySQL relational database to ensure proper organization and accessibility.

B. Data Processing Scheme and Algorithms

The system processes data through several stages, including pre-processing and post-processing, to ensure accuracy, security, and efficiency. The raw dataset initially consists of unvalidated user inputs from registration forms and login credentials, product entries such as names, prices, stock levels, and image links, as well as order details including selected items and quantities. Before storing or using the data, pre-processing steps are applied. These include input validation to ensure that all required fields are properly filled and formatted, data sanitization to prevent security threats such as SQL injection, and password encryption to securely store user credentials. Additionally, data formatting is performed to standardize values such as currency, text capitalization, and date formats.

After pre-processing, the dataset becomes clean, structured, and secure, making it ready for system use. Post-processing steps are then applied, including inventory synchronization, where stock levels are automatically updated after each successful transaction, and order processing logic, which calculates total costs, verifies product availability, and records transactions. The processed data is then retrieved and displayed dynamically on the website through the integration of Java for backend processing, HTML and CSS for the frontend interface, and MySQL for database management.

C. Other Data Utilized in the Design

In addition to the main datasets, the system also utilizes other supporting data that are essential for its overall functionality. This includes administrator credentials used for managing products, users, and orders within the system. System constants, such as shipping fees, tax rates, and default configuration settings, are also incorporated to maintain consistency in calculations and operations. Moreover, mock data, including sample product listings and test user accounts, are used during the development and testing phases to ensure the system functions correctly. Temporary session data is also utilized to manage user login states and maintain shopping cart information during user interaction with the website.

2.3 Design Alternatives

The various design options taken into consideration for the Clothing Brand Web System for Online Retail Operations are shown in this section. Each option offers a workable way to put the system into place while meeting the project's goals and limitations in terms of cost, performance, security, scalability, and usability. To make sure that the final design choice is technically sound, economical, and suitable for the client's business's operating size, it is essential to evaluate several options.

2.3.1 Rationale for Design Alternatives

The chosen design options concentrate on several web system architectures that are frequently employed in e-commerce platforms. These options were selected because they have a direct impact on system performance, security, scalability, development complexity, and maintenance costs—all of which are crucial limitations mentioned in Chapter 1.

The design team was able to assess how various structural approaches affect system behavior under real-world circumstances like concurrent user access, secure transaction handling, and budget constraints by contrasting a monolithic architecture, a microservices-based architecture, and a serverless/cloud-based architecture. For a small to medium-sized clothing firm, these options are important because they direct the decision-making process toward a design that strikes a balance between technical capacity and practical viability.

2.3.1 Design Alternative 1:

A. Engineering Principles of Alternative

Tight component integration, centralized system architecture, and process simplicity are the tenets of this design option. A single backend program is used to implement all system functions, including order processing, inventory control, product administration, and user authentication. Internal communication between components minimizes deployment overhead and architectural complexity.

This method is appropriate for small-scale systems with tight budgets because it places an emphasis on simplicity of development, simple debugging, and low infrastructure requirements.
Note: This repeats for all 3 designs. It must not mention the principles mentioned above but instead those specific to the design alternative.

B. Architecture of Design Alternative

The system is set up as a single web application that includes:

- A React-based front end for communicating with administrators and customers
  - All business logic is handled by an Express and Node.js backend.
  - Users, goods, orders, and inventory information are all stored in a single database.

The same server program handles all client requests; it talks with the database directly and provides answers to the front end.

C. Constraints

- **Constraint A (Performance):**
  Performance constraint relates to the speed and responsiveness of the website, measured by load and response time. Poor performance reduces user satisfaction. Therefore, faster systems are more effective. Hence, the design with the best performance is considered the best design.

- **Constraint B (Security):**
  Security constraint relates to protecting user data and system integrity through secure methods. Weak security increases the risk of data breaches. Therefore, stronger security is more reliable. Hence, the design with the highest security is considered the best design.

- **Constraint C (Scalability):**

Scalability constraint relates to the system's ability to handle growth in users and data. Poor scalability leads to slowdowns or failures. Therefore, scalable systems are more reliable. Hence, the design with the highest scalability is considered the best design.

- **Constraint D (Efficiency):**
  Efficiency constraint relates to the optimal use of resources such as time, cost, and system power. Inefficiency increases costs and slows performance. Therefore, efficient systems are more effective. Hence, the design with the highest efficiency is considered the best design.

- **Constraint E (Technology):**
  Technology constraint relates to the tools and platforms used in development. Poor technology choices can limit performance and maintenance. Therefore, modern and compatible technologies are more effective. Hence, the design with the best technology is considered the best design.

2.3.2 Design Alternative B

I. Engineering Principles of Alternative

This alternative is based on distributed systems, service independence, and loose coupling. Each major function of the system—authentication, product catalog, inventory management, and order processing—is implemented as a separate service with its own logic and interfaces.

This approach improves scalability and fault isolation by allowing services to operate and scale independently.

II. Architecture of Design Alternative

The system is divided into multiple backend services, each exposed through RESTful APIs:

- A front-end React application communicates with multiple backend services
- Each service runs on Node.js and handles a specific function
- Services interact with a shared or service-specific database layer

This architecture allows individual services to be updated or scaled without affecting the entire system.

III. Evaluation of Constraints

- **Constraint A (Performance):** Improved performance under heavy load but with added network communication overhead.
- **Constraint B (Security):** Better isolation of sensitive services, though increased complexity in securing inter-service communication.
- **Constraint C (Scalability):** High scalability due to independent service deployment.

2.3.3 Design Alternative C

I. Engineering Principles of Alternative

This alternative applies event-driven computing, cloud resource abstraction, and automatic scalability. Backend logic is implemented using cloud-based serverless functions that execute in response to user requests, eliminating the need for dedicated server management.
The system relies heavily on managed cloud services for computing, authentication, and database storage.

II. Architecture of Design Alternative

- A React front end communicates with cloud-hosted APIs
- Backend logic is executed through serverless functions
- A managed cloud database stores system data
- The cloud provider automatically handles scaling, availability, and infrastructure maintenance

III. Constraints

- Performance Constraints (Response Time): Good performance after initialization, but may experience cold-start delays.

- Security Constraints (Data Protection Level): Strong built-in security features from cloud providers.

- Scalability Constraints (Number of Concurrent Users): Excellent scalability, automatically adjusting to user demand

2.4 Standards Involved in the Design

This section presents the standards followed by the design, including their references. Matrix may be used to show how standards are used in each specific design.

Table 2.4 Summary of Standards Involved in the Alternatives

Standard	Brief Description	DESIGNS		
		DESIGN A	DESIGN B	DESIGN C
WCAG 2.1 Level AA (Web Accessibility Standard)	Web accessibility standard ensuring usability for users with disabilities.	Used to ensure readable text, proper contrast, and alt-text for product images.	Used to support keyboard navigation and accessible checkout process.	Used to ensure accessibility features remain consistent and functional as the system scales across different devices and increasing users.
OWASP Top 10 (Web Security Standard)	Web security standard identifying common web application vulnerabilities.	Used to secure user authentication and form inputs.	Used to protect payment and transaction processes.	Used to prevent data exposure and injection attacks.
ISO/IEC 25010 (Software Quality Standard)	Software quality standard for evaluating system performance and reliability.	Used to verify functional correctness of core features.	Used to assess system performance and responsiveness.	Used to evaluate usability and security during testing.
ISO/IEC 27001 (Information Security Standard)	Information security management standard.	Used to secure customer accounts and personal data.	Used to protect order and payment records.	Used to control administrator access and database security.
ISO/IEC 14543-3 (System Architecture Standard)	Standard supporting structured system architecture and component interaction. system components.	Used as reference for front-end and back-end integration.	Used to guide communication between system modules.	Used to support scalable system architecture, allowing efficient expansion of system components and services.

Explain of the Table

The table summarizes the standards considered in the design of the online clothing brand system and illustrates how each standard is applied across different design alternatives. WCAG 2.1 Level AA is used to enhance accessibility and usability for all users. OWASP Top 10 focuses on identifying and mitigating common security threats to protect user data and transactions. ISO/IEC 25010 ensures the system meets quality expectations such as reliability, performance, and usability. ISO/IEC 27001 is applied to safeguard sensitive information, while ISO/IEC 14543-3 supports the organization and interaction of system components.

Summary

In summary, applying these standards ensures that the online clothing brand system is accessible, secure, reliable, and well structured. The standards guide the design choices across different alternatives, helping the system meet user requirements, security constraints, and overall project objectives.

CHAPTER 3: DESIGN TRADEOFFS

3.1 Summary of Constraints

The three design alternatives will be evaluated by comparing them against the identified constraints performance, security, scalability, efficiency, and technology to determine their effectiveness in addressing the problem and meeting the project objectives and standards. The table below presents the comparison of the three designs, highlighting key factors such as system speed and responsiveness (performance), data protection (security), ability to handle expected user load (scalability), optimal use of resources (efficiency), and the use of appropriate and modern tools (technology).

Table 1 Summary of Design Constraints

Designs	**Constraints**				
	Performance (Page Load Time)	**Security (User Protection)**	**Scalability (User Capacity)**	**Efficiency (Resource Usage)**	**Technology (Tools & Platforms)**
Design 1	1.2 seconds (Fast)	High (Secure login)	Up 15 users	Low resource usage	Modern & compatible tools (Html,CSS, JS,PHP)
Design 2	1.8 seconds (Moderate)	Moderate (Login system only)	10-12 users	Moderate resource usage	Standard web technologies
Design 3	2.5 second (Slow)	Basic security	Up to 10 users	High resource usage	Less optimized tools

The table presents a comparison of three design alternatives for a clothing shop website based on performance, security, scalability, efficiency, and technology. Among the three, Design 1 demonstrates the best overall performance, offering faster page load time, stronger security features, better scalability for expected users, and more efficient use of resources. Additionally, it utilizes modern and compatible technologies, making it more reliable and suitable for the system requirements. Therefore, Design 1 is considered the most effective solution in meeting the project objectives and standards.

3.2 Trade-offs

Table 2 Preference and Importance of Constraints

Constraints	Preference	Raw Importance	% Importance
Performance (Page Load Time)	Minimization	10	25%
Security (Data Protection)	Maximization	9	22.50%
Scalability (User Capacity)	Maximization	8	20%
Efficiency (Resource Usage)	Minimization	7	17.50%
Technology (Tools & Platforms)	Maximization	6	15%
Total		40	**100%**

Explain the use of Pareto Multi-Criteria Decision Making (MCDM).

$$Minimization = 9 \times \left(\frac{Max\,Value - Raw\,Value}{Max\,Value - Min\,Value}\right) + 1$$ Equation No. xx

$$Maximization = 9 \times \left(\frac{Raw\,Value - Min\,Value}{Max\,Value - Min\,Value}\right) + 1$$ Equation No. xx

3.2.1 Trade-off 1: Performance (Page Load Time)

The performance constraint is evaluated based on the page load time of the website. This measures how quickly the clothing shop system responds to user actions such as browsing products and adding items to the cart. Among the three designs, Design 1 shows the fastest response time, making it the most efficient in terms of performance, while the other designs exhibit slower loading speeds.

Table 3.2.1.1 Evaluation of Three Design Alternatives based on Performance

Design	Performance (Page Load Time)
Design 1	1.2 seconds
Design 2	1.8 seconds
Design 3	2.5 seconds

3.2.1.2 Normalization Values of Design for Performance

Design	Normalized Value
Design 1	10
Design 2	5.85
Design 3	1

$$= \frac{70 - 40}{100 - 40} \times 9 + 1$$

$$= \frac{30}{60} \times 9 + 1$$

$$= 0.5 \times 9 + 1 = 4.5 + 1 = 5.5 \approx 5.85$$

Table 3.2.1.2 shows the normalized values of the three designs in terms of Performance. Design 1 obtained the highest normalized value of 10.00, indicating that it has the best system performance among the alternatives. Design 2 has a normalized value of 5.85, showing moderate performance. Lastly, Design 3 has the lowest normalized value of 1.00, indicating the least efficient performance among the three designs.

3.2.1.3 Evaluation of Three Design Alternatives based on Security

Design	Security (Level)
Design 1	High
Design 2	Moderate
Design 3	Basic

3.2.2 Tradeoff 2: Constraint B (Metric)

3.2.2.1 Normalization Values of Design for Security (maximization: Higher is better)

Design	Normalized Value
Design 1	10
Design 2	5.50
Design 3	1

Table 3.2.2.1 presents the normalized values of the three designs based on Security. The normalization was derived using a direct rating scale, where qualitative levels were converted into numerical values. Design 1, which has a high security level, was assigned the highest normalized value of 10.00. Design 2, with a Moderate level of security, was assigned a normalized value of 5.50. Lastly, Design 3, having a Basic level of security, received the lowest normalized value of 1.00.

$$= \frac{2-1}{3-1} \times 9 + 1$$

$$= \frac{1}{2} \times 9 + 1$$

$$= 4.5 + 1 = 5.5$$

3.2.2.2 Evaluation of Three Design Alternatives based on Scalability

Design	Scalability (users)
Design 1	15 users
Design 2	12 users
Design 3	10 users

Best: Design A

3.2.2.3

3.2.3 Tradeoff 3: Constraint C (Metric)

3.2.3.1 Design 1: Normalization of Constraint C (Metric)
3.2.3.2 Design 2: Normalization of Constraint C (Metric)
3.2.3.3 Design 3: Normalization of Constraint C (Metric)
3.2.4 Tradeoff 4: Constraint D (Metric)

3.2.4.1 Design 1: Normalization of Constraint D (Metric)
3.2.4.2 Design 2: Normalization of Constraint D (Metric)
3.2.4.3 Design 3: Normalization of Constraint D (Metric)
3.2.5 Tradeoff 5: Constraint E (Metric)

3.2.5.1 Design 1: Normalization of Constraint E (Metric)
3.2.5.2 Design 2: Normalization of Constraint E (Metric)
3.2.5.3 Design 3: Normalization of Constraint E (Metric)

3.3 Summary of the Normalized Values of the Three Designs

Designs	Constraints				
	Constraint A (metric)	Constraint B (metric)	Constraint C (metric)	Constraint D (metric)	Constraint E (metric)
Design A					
Design B					
Design C					

3.4 Designers Raw Ranking for the Three Designs

Table xx Designers Raw Ranking for the Three Designs

Decision Criteria	Criterion's Importance		Ability to Satisfy Criterion		
	Scale (0-10)	Percentage (%)	Design A	Design B	Design C

3.5 Sensitivity Analysis
3.6 Influence of the Design Tradeoffs in the Final Design

CHAPTER 4: FINAL DESIGN

4.1 Final Design
4.1.1 Software Application
4.1.2 Hardware Design
4.2 Test Procedures and Evaluation
4.2.1 Test Procedures
4.2.2 Test Evaluation
4.3 Test and Evaluation Results
4.3.1 Test Results
4.3.2 Evaluation Results
4.4 Conclusion
4.5 Impact of the Design
4.5.1 Societal
Target UN SDG.
4.5.2 Ethical
In compliance with known ethical codes/standards
4.5.3 Legal
National / Intl Laws
4.6 Sustainability Plan

CHAPTER 5: BUSINESS PLAN AND MODEL

5.1 Business Plan
5.1.1 Executive Summary
5.1.2 General Company Description
5.1.3 Products and Services Offered
5.1.4 Marketing Plan
5.1.5 Marketing Strategy
5.2 Business Model
5.3 Intellectual Property (IP) Reports

REFERENCES

Note: This must be done using APA format. Check the guide for more details: https://www.scribbr.com/apa-style/apa-seventh-edition-changes/

Covey, S. R. (2013). *The 7 habits of highly effective people: Powerful lessons in personal change*. Simon & Schuster.

APPENDICES

Include standards preview, certification from experts/clients, code snippets, patent reports, and other long and detailed documents here. Format is as follows below:

APPENDIX A: TITLE OF THE SECTION

<figure>

Note: No figure number. Standards must be followed with a paragraph explaining its contents and purpose.